The Balanced Readout

Why the winner-take-all territory diverges from the constrained mass, and the dual-offset correction that removes the divergence

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Abstract

Phase 1 minimises a Γ-convergence energy over continuous densities and enforces equal area as a constraint on the *continuous masses*. The partition it delivers, however, is the *winner-take-all territory* of those densities. This document derives the exact identity relating the two, shows why nothing in the variational problem controls their difference, and analyses the two artifacts that follow — territory far from the equal-area target, and territory that splits into disconnected pieces. It then derives the correction implemented in `src/partition/balanced_readout.py`: a per-cell additive offset obtained from the dual of a transportation problem. Three properties are proved: the corrected readout grows each cell *monotonically outward from its own core* (the locality that in-flow corrections lacked); the attainable balance is bounded below by the one-vertex granularity, and this bound is a theorem rather than an observation, because the underlying linear program is *not* integral; and the offsets control area but not connectivity, which is why a second, connectivity-preserving pass is required. Where an argument is a model rather than a derivation — notably the worst-cell scaling in N, and the mechanism behind disconnection — this is stated explicitly.

Contents

1 Overview

The method of Bogosel and Oudet [3] represents a partition of a surface into N cells by N density functions and minimises a Modica–Mortola-type functional [9, 8] whose Γ-limit is the total interface length. Equal areas are imposed as constraints on the integrals of those densities.

To hand a partition downstream one must convert the densities into a definite assignment of each mesh vertex to one cell. This step is **prescribed by the source paper itself**, whose equation (5–1) defines

$$\phi_i(x) = \begin{cases} 1 & \text{if } u_i(x) \geq \max_{j \neq i} u_j(x), \\ 0 & \text{otherwise,} \end{cases}$$

i.e. pointwise *winner-take-all*. We refer to this conversion as the *readout*, by analogy with instrumentation, where the internal state of a device and the value it reports are distinct objects and the conversion between them is a separate source of error.

Remark 1.1 (What the source paper does and does not say about its own readout)**.** Read directly (`docs/papers/Partitions of Minimal Length on Manifolds.pdf`), [3] *does* discuss the readout and names two difficulties with it: a small void around each triple point, "included in one of the triangles of the mesh, and can be dealt with"; and contours that "pass through the middle of the edges", producing zigzagged boundaries "whose length is significantly larger than the optimal total perimeter". The second is the stated motivation for the direct constrained optimization that follows — our Phase 2 — which re-imposes equal area *on the discrete cells*.

Neither named difficulty is the one analysed here. The paper's reported "Area tol." figures, 2–5 $\times 10^{-7}$, are the residual of the *continuous* constraint $\langle v, \phi_i \rangle = A/n$ — the same constraint we use — and not the deviation of the *extracted* territory. That latter quantity is what (5) below computes, and it does not appear in the paper.

The reason is scale, and it is entirely reasonable. The paper demonstrates $n \in \{2, \ldots, 24, 32\}$ on the sphere, $n \in \{2, 4, 6\}$ on the double torus, and — on the torus with exactly our parameters $R = 1$, $r = 0.6$ — $n \in [2, 11]$. This project works at $N = 100$ to 400, i.e. **roughly 10 to 36$\times$ beyond the largest case the method was demonstrated on for this surface.** By (7) the readout error grows like $\sqrt{N/V}$, so at $n \leq 11$ it is small enough that the subsequent constrained optimization absorbs it, and the question never arises. The contribution here is therefore a *scaling* result, not the discovery of an oversight.

One further observation, recorded neutrally: the paper notes that a cell's discrete contour "may have one or more connected components". That is a statement about the contour data structure accommodating several loops, not an analysis of cells fragmenting as a failure mode, and it should not be read as anticipating §5.

The central observation of this document is that the constrained quantity and the delivered quantity are *different functionals of the same field*, and that their difference is exactly computable, is supported entirely on the diffuse interface band, and is referenced nowhere in the energy or the constraints. It is therefore free to drift, and it drifts further as the number of cells grows at fixed mesh resolution.

Remark 1.2. This is a statement about the readout at finite ε and finite mesh. The Γ-convergence result concerns the limit of the *energy* and is not in question. Nothing here contradicts [3]: its readout is well suited to the range it was demonstrated on, and what follows quantifies an error term that only becomes binding an order of magnitude beyond that range.

Related documents: `docs/math/06-phase1-energy-discretization/` derives the discrete energy itself; `docs/math/07-phase1-wta-balance/` derives a *rejected* attempt to enforce balance inside the flow, and its Proposition 1 is the identity restated as (5) below; `docs/reference/winner_take_all_partition_gap.md` is the standing prose account.

2 Notation and Setup

Let the surface be triangulated with vertex set $\{1, \ldots, V\}$ and let M be the P1 finite-element mass matrix. Write

$$v_i \;=\; \sum_j M_{ij} \tag{1}$$

for the lumped mass of vertex i, so that $v_i > 0$ and $\sum_i v_i = |\Sigma|$, the total surface area. The densities are $u_{ik} \in [0, 1]$ for vertex i and cell $k \in \{1, \ldots, N\}$, subject to the partition-of-unity and equal-area constraints

$$\sum_{k=1}^{N} u_{ik} \;=\; 1 \quad \forall i, \qquad \sum_{i=1}^{V} v_i\, u_{ik} \;=\; \bar{A} \;=\; \frac{|\Sigma|}{N} \quad \forall k. \tag{2}$$

Definition 2.1 (Readout and territory)**.** The *readout* is the map $\omega : \{1, \ldots, V\} \to \{1, \ldots, N\}$,

$$\omega(i) \;=\; \arg\max_k \, u_{ik}, \tag{3}$$

ties broken by lowest index. The *territory* of cell k is $\Omega_k = \{i : \omega(i) = k\}$ and its *discrete area* is

$$T_k \;=\; \sum_{i \in \Omega_k} v_i. \tag{4}$$

The constraint in (2) fixes $\sum_i v_i u_{ik}$; validity is graded on T_k. These coincide only if every u_{ik} is exactly 0 or 1.

3 The readout gap is exact

Proposition 3.1 (Territory–mass identity)**.** *For every cell k, with the constraint (2) satisfied,*

$$T_k - \bar{A} \;=\; \underbrace{\sum_{i \in \Omega_k} v_i\,(1 - u_{ik})}_{\mathrm{gain}_k} \;-\; \underbrace{\sum_{i \notin \Omega_k} v_i\, u_{ik}}_{\mathrm{lost}_k}. \tag{5}$$

Proof. Split the constrained sum over the territory and its complement:

$$\bar{A} \;=\; \sum_i v_i u_{ik} \;=\; \sum_{i \in \Omega_k} v_i u_{ik} \;+\; \sum_{i \notin \Omega_k} v_i u_{ik} \;=\; \Big(T_k - \sum_{i \in \Omega_k} v_i (1 - u_{ik})\Big) \;+\; \mathrm{lost}_k,$$

using $\sum_{i \in \Omega_k} v_i u_{ik} = T_k - \sum_{i \in \Omega_k} v_i (1 - u_{ik})$ from (4). Rearranging gives (5). □

Remark 3.1 (The unconstrained form, and when it matters)**.** The proof uses the constraint only in the last step. Unconditionally,

$$T_k - \sum_i v_i u_{ik} \;=\; \mathrm{gain}_k - \mathrm{lost}_k,$$

so if the equal-area constraint holds only to a residual $\delta_k = \sum_i v_i u_{ik} - \bar{A}$, then $T_k - \bar{A} = \mathrm{gain}_k - \mathrm{lost}_k + \delta_k$. With the exact projection $\delta_k \sim 10^{-10}$ and the distinction is immaterial. It is *not* immaterial for the soft-area variant, which replaces the constraint by a penalty and therefore leaves δ_k uncontrolled — and that variant is precisely the one whose viability depends on this readout.

The identity is exact: no linearisation, no assumption on ε or on the mesh. Its two terms have a direct reading. gain_k is mass that cell k *owns but does not hold* — vertices it wins while its density there is below one. lost_k is mass it *holds but does not own* — density deposited at vertices won by somebody else.

Remark 3.2 (The gap lives entirely in the interface band)**.** Both terms vanish wherever the field is crisp. Deep inside cell k, $u_{ik} \approx 1$, so $1 - u_{ik} \approx 0$ and the vertex contributes nothing to gain_k. Far outside, $u_{ik} \approx 0$ and it contributes nothing to lost_k. Only vertices in the diffuse band, where $0 < u_{ik} < 1$, contribute to either. *The readout gap is a band phenomenon*, and everything about how it scales follows from how much of a cell the band occupies.

3.1 Why nothing controls it

The energy is a function of u; the constraints (2) are linear in u. Neither references ω. Moreover the map $u \mapsto \omega$ is piecewise constant and therefore discontinuous: it is constant on each cell of the arrangement $\{u_{ik} - u_{il} = 0\}$ and jumps across it. Consequently

- T_k has zero gradient with respect to u almost everywhere, so no first-order optimality condition can constrain it;
- adding T_k to the objective is not possible without a surrogate, and the surrogate used in `docs/math/07-phase1-wta-balance/` was measured and rejected (see §6.3);
- the quantity $\mathrm{gain}_k - \mathrm{lost}_k$ is, from the optimiser's point of view, an unobserved by-product.

4 Artifact I: territory far from the equal-area target

4.1 The band fraction

Let $R_{\mathrm{cell}} = \sqrt{|\Sigma|/(\pi N)}$ be the radius of a disc of the target area and let h be a representative edge length. In the discretisation used here the interface width parameter is tied to the mesh, $\varepsilon \sim h$ (`docs/math/06-phase1-energy-discretization/`). A cell's band is a collar of width $O(\varepsilon)$ around a boundary of length $O(R_{\mathrm{cell}})$, so the fraction of a cell occupied by band is

$$f_b \sim \frac{\varepsilon \cdot 2\pi R_{\mathrm{cell}}}{\pi R_{\mathrm{cell}}^2} = \frac{2\varepsilon}{R_{\mathrm{cell}}} \sim \frac{h}{R_{\mathrm{cell}}}. \tag{6}$$

With $h \sim \sqrt{|\Sigma|/V}$ this becomes

$$f_b \sim \sqrt{\frac{N}{V}} = \left(\frac{V}{N}\right)^{-1/2}, \tag{7}$$

i.e. **the band fraction is governed by vertices per cell alone.** This recovers, from geometry, the empirical form $f_b = 3.72/\sqrt{2V/N}$ recorded in `docs/reference/winner_take_all_partition_gap.md`.

Remark 4.1 (The scaling trap)**.** At *fixed mesh*, raising N shrinks R_{cell} and therefore *raises* f_b: the band becomes a larger share of each cell precisely as the cells become the objects one cares about. Holding f_b fixed requires $V \propto N$ — constant vertices per cell — so the mesh must grow linearly in the number of cells merely to keep the readout error from worsening.

4.2 The diffuse runt

Equation (5) permits a configuration that is feasible, low-energy, and useless: a cell whose density is spread as a broad halo, everywhere slightly below its neighbours'. It then satisfies $\sum_i v_i u_{ik} = \bar{A}$ exactly while winning almost nothing, because lost_k is large and gain_k small. Such a cell is *confidently in the right place and insufficiently loud.* Empirically at $N = 300$ the worst cell reached 36.15% below target while its continuous mass was exact.

4.3 Worst-of-N — a model, not a derivation

Remark 4.2 (Explicitly not established)**.** The following is the argument by which the project has explained why the problem sharpens with N. It is a **model**. The distribution of $\text{gain}_k - \text{lost}_k$ across cells has never been measured, and the available data points differ in both λ and mesh, so they cannot test it.

If one models each cell's net exchange as a random variable with mean zero and standard deviation $\sigma \propto f_b$, roughly independent across cells, then validity — which is graded on $\max_k |T_k - \bar{A}|$ — is an extreme-value statistic, and for light-tailed exchange the maximum of N draws grows like $\sigma\sqrt{2\ln N}$. Two consequences follow *if the model holds*: the worst cell degrades slowly but without bound as N grows even at fixed f_b; and any mechanism that merely *reduces the variance* of the exchange postpones the problem rather than removing it. Only a mechanism that makes the delivered representation balanced *by construction* is immune to the tail. That prediction is consistent with every measured outcome in this project, which is evidence for the model but not a proof of it.

5 Artifact II: disconnected territory

Remark 5.1 (Weaker than §4, and deliberately so)**.** The existence of this artifact is a theorem (§5.1); its *cause* in our runs is a hypothesis that has never been traced to a specific event. The two are separated below.

5.1 Nothing forbids it

The readout (3) is defined pointwise. Ω_k is a super-level set of the function $i \mapsto u_{ik} - \max_{l\neq k} u_{il}$, and a super-level set of an arbitrary function on a graph need not be connected. Neither the energy nor the constraints contain any term that is finite for connected territories and infinite otherwise. Disconnection is therefore *admissible* and costs nothing directly; a split cell is a legitimate local minimiser of the relaxed problem.

What makes it pathological is external: a minimal-perimeter cell is connected, so a split territory is a non-physical artifact of the representation, and Phase 2 cannot consume it, since a split cell yields multi-loop contours.

Remark 5.2 (Why the other two gates are blind to it)**.** A split cell's pieces sum to the correct area and each piece is crisp. It therefore passes a dormant-cell check (peak density ≈ 1) and an area-imbalance check (total area correct). Connectivity is a third, independent validity dimension.

5.2 The mechanism — hypothesis

The constraint $\sum_i v_i u_{ik} = \bar{A}$ is enforced by an iterative projection onto the feasible set. The projection is a *nonlocal* operator: correcting a deficit in cell k adds a rank-one term spread across the whole free set, followed by renormalisation. Raising u_{ik} everywhere by a near-uniform amount can therefore flip the argmax at vertices arbitrarily far from the cell's core — wherever cell k happened to be a close second — and open a second basin.

Three observations are consistent with this and none establishes it:

1. Splits appear *while balance is improving.* At $N = 300$ they are born mid-level-2, at the same time as the imbalanced count falls from 234 to 10: balance is being bought with disconnection.

2. Mechanisms that push balance through the projection make it dramatically worse. A discrete-area trim produced 14/200 split cells against a matched control's 0/200; a soft area constraint produced 3–4/100.

3. The correction of §6 does not *leave* splits — but, contrary to what an earlier draft of this document claimed, it does not avoid creating them either. See Remark 6.4.

Remark 5.3 (What would settle it)**.** Decompose the per-iteration update into its gradient and projection contributions at the iteration a split is born, and test whether the new component's vertices had u_{ik} raised by the projection. This has not been done; it is recorded as future work in `docs/plans/PUBLICATION_READINESS_PLAN.md`.

6 The corrected readout

6.1 The assignment problem

Fix any per-vertex score matrix s_{ik}; the shipped readout uses $s_{ik} = \log u_{ik}$, with densities clipped from below so that exact zeros give a finite, very negative score. We seek a hard assignment maximising total score subject to every cell attaining its target area:

$$\max_x \sum_{i,k} v_i\, s_{ik}\, x_{ik} \quad \text{s.t.} \quad \sum_k x_{ik} = 1 \ \ \forall i, \qquad \sum_i v_i\, x_{ik} = \bar{A} \ \ \forall k, \qquad x_{ik} \geq 0. \tag{8}$$

This is a transportation problem: supplies v_i at vertices, demands $\bar{A}$ at cells, and both marginals sum to $|\Sigma|$, so it is feasible.

Remark 6.1 (Terminology)**.** This project's prose has called (8) a *semi-discrete* optimal transport problem. Strictly it is *discrete-to-discrete*: V weighted points to N cells. The semi-discrete problem — a continuous source measure transported to N atoms, in the sense of [2, 7, 6] — is its continuum limit as the mesh refines; see [10, §5] for the distinction. The offsets derived below are the discrete analogue of the power weights in that literature.

6.2 The dual, and where the offsets come from

Introduce multipliers φ_i for the row constraints and ψ_k for the column constraints; this is Kantorovich duality for a finite transportation problem [11], and the construction and its solvers are set out in [10]. Stationarity of the Lagrangian in x_{ik} gives $v_i s_{ik} - \varphi_i - v_i\psi_k \leq 0$, with equality wherever $x_{ik} > 0$, so $\varphi_i = v_i \max_k(s_{ik} - \psi_k)$ and the dual is

$$\min_{\psi\in\mathbb{R}^N} D(\psi), \qquad D(\psi) \;=\; \sum_i v_i \max_k \big(s_{ik} - \psi_k\big) \;+\; \bar{A}\sum_k \psi_k. \tag{9}$$

D is a sum of pointwise maxima of affine functions plus a linear term, hence **convex** and piecewise linear. Complementary slackness forces every vertex onto a maximising cell, so an optimal assignment is

$$\omega_\psi(i) \;=\; \arg\max_k \big(s_{ik} - \psi_k\big). \tag{10}$$

Remark 6.2 (Sign convention)**.** The implementation stores $\psi^{\text{code}} = -\psi$ and evaluates $\arg\max_k(s_{ik} + \psi_k^{\text{code}})$, in which *raising* ψ_k *grows* cell k. (9), (11) and Proposition 6.1 are stated in the *math* convention $(s - \psi)$, where the derivative is $\bar{A} - T_k$; (12) and §6.3 use the code convention, where the corresponding quantity is $T_k - \bar{A}$. Both describe the same descent — grow the cells that are too small — and the implemented update is descent in its own convention.

Proposition 6.1 (The implemented update is the dual subgradient)**.** *At any ψ where the argmax is unique at every vertex, D is differentiable with*

$$\frac{\partial D}{\partial \psi_k} \;=\; \bar{A} - T_k(\psi), \qquad T_k(\psi) = \sum_{i\,:\,\omega_\psi(i)=k} v_i, \tag{11}$$

and in general this is a subgradient. Hence gradient descent on D is exactly "grow the cells that are too small, shrink the cells that are too large".

Proof. Differentiating the first term of (9) at a point of uniqueness picks out, for each i, the maximising index $\omega_\psi(i)$, contributing $-v_i$ to the ψ_k derivative precisely when $\omega_\psi(i) = k$; the second term contributes $\bar{A}$. Summing gives (11). At points where the argmax is non-unique, D is a maximum of finitely many affine functions and the same expression is an element of the subdifferential for any measurable tie-breaking. □

Code: dual ascent on the offsets

`src/partition/balanced_readout.py: solve_dual_offsets(scores, lumped_mass, target, config)`

`psi -= (eta0 / (1 + decay*it)) * rel`, with `rel = (areas - target)/target` — the normalised form of (11). The routine returns its *best* iterate, since D is piecewise linear and subgradient descent is not monotone. Scores enter only through `argmax(scores + psi)`, so the routine is correct for any additive score matrix; only its step schedule is scale-dependent.

Remark 6.3 (Scope of the LP discussion)**.** Everything said about (8) concerns its *optimum*. The readout in practice uses the best iterate of a subgradient method, which need not be optimal; and the family $\{\omega_\psi\}$ explored is only the N-parameter shifted-argmax subfamily of all N^V assignments. Both restrictions are deliberate — the subfamily is what makes the correction local — but neither is a consequence of the LP.

6.3 Locality: cells grow outward from their own cores

Proposition 6.2 (Nested monotone growth)**.** *Fix ψ_{-k} and regard $\Omega_k(\psi_k)$ as a function of the single offset ψ_k. Define the* margin *of vertex i against cell k as*

$$m_i \;=\; \max_{l \neq k} \left(s_{il} + \psi_l\right) \;-\; s_{ik}. \tag{12}$$

Then $\Omega_k(\psi_k) = \{i : m_i \leq \psi_k\}$. Consequently $\psi_k < \psi'_k$ implies $\Omega_k(\psi_k) \subseteq \Omega_k(\psi'_k)$, and vertices are acquired in nondecreasing order of m_i.

Proof. Immediate from (10): $i \in \Omega_k$ iff $s_{ik} + \psi_k \geq \max_{l \neq k}(s_{il} + \psi_l)$, i.e. iff $\psi_k \geq m_i$. The sets are sublevel sets of the fixed function m, hence nested. □

This is the property the in-flow corrections lacked, and it is worth stating plainly what it does and does not say.

- **What it gives.** Growth is *ordered by margin.* A cell claims first the vertices where it was most nearly winning, which — for a density field that is unimodal about the cell's core — are the vertices adjacent to the territory it already holds. Correction proceeds outward through the adjacent band rather than being deposited wherever the algebra puts it.

- **What it does not give.** Proposition 6.2 is a statement about the *order* of acquisition, not about connectivity. If the margin function m has two separated basins — exactly the situation Artifact II describes — then nested growth will populate both. Locality prevents the correction from *manufacturing* new components; it cannot repair components that the density field already possesses. Hence §7.

Remark 6.4 (Measured: the offsets increased fragmentation before repair removed it)**.** Proposition 6.2 is comparative statics in a *single* coordinate: it fixes ψ_{-k}. The algorithm moves all N offsets jointly, so neither the trajectory nor the net change from the source labelling to the shifted labelling is nested, and no conclusion about connectivity follows.

The recorded data confirm this concretely. In the $N = 300$ run whose figures appear in §9 (`run_20260806_123326`), the two fragmented cells go from **2 components each** in the source labelling to **4 components each** after the dual shifts, and cell 290's largest stray grows from

0.0032 to 0.0183 — from below to well above the 1%-of-$\bar{A}$ reporting threshold. The per-cell fragmented *count* is unchanged at 2, which is why the summary figures do not show it. Repair then removes all of it.

So the honest statement is: the offsets correct area, they can worsen connectivity en route, and the repair stage is not a tidying-up step but a necessary component. An earlier draft asserted the opposite from Proposition 6.2 alone, without consulting the metadata of the very run it was quoting.

Contrast this with enforcing balance inside the flow. There the correction reaches the field through the constraint projection, which is nonlocal (§5); measured outcomes were a net regression in both attempts (`docs/experiments/04-territory-aware-highn-validation/`, `docs/experiments/05-soft-area-constra`

6.4 Exact balance is generically unattainable; the vertex mass is a scale, not a floor

It is tempting to assert that (8) has an integral optimum because transportation problems do. That reasoning is unavailable here.

Proposition 6.3 (Non-integrality)**.** *The classical integrality theorem for the transportation polytope [1, Ch. 9] requires integer supplies and demands. In (8) the supplies are the real lumped masses v_i and the demands are $\bar{A} = |\Sigma|/N$. Unless some subset $S \subseteq \{1, \dots, V\}$ satisfies $\sum_{i \in S} v_i = \bar{A}$ exactly, no integral feasible point exists at all, and every vertex of the polytope splits at least one vertex's mass between cells. A basic feasible solution of a transportation problem with V sources and N sinks corresponds to a spanning forest of the bipartite graph and so has at most $V + N - 1$ nonzeros [1, Ch. 11], hence at most $N - 1$ split vertices.*

Consequently the exact optimum of (8) is in general *fractional*, the readout (10) is a rounding of it, and rounding the $\leq N - 1$ split vertices perturbs the affected cells by at most $\max_i v_i$ each. Since exact subset-sum coincidences do not occur for generic masses, it follows that

$$\min_{\text{hard assignments}} \; \max_k \left| T_k - \bar{A} \right| \; > \; 0. \tag{13}$$

Remark 6.5 (What this does *not* say — a corrected claim)**.** An earlier version of this document asserted the stronger statement $\min \max_k |T_k - \bar{A}|/\bar{A} \geq \max_i v_i/\bar{A}$, and called it a theorem on the strength of Proposition 6.3. **That is a non-sequitur and it is false.** Rounding a split vertex costs *at most* $\max_i v_i$ — an upper bound on the damage — which says nothing about how small the best achievable deviation is. A counterexample settles it: take $V = 7$, $N = 2$, masses $(4, 1, 1, 1, 1, 1, 1)$ perturbed generically so that no exact subset sum exists, so $\bar{A} = 5$ and $\max_i v_i/\bar{A} = 80\%$; the assignment placing the heavy vertex with one light vertex against the remaining five achieves a worst deviation of $3.4 \times 10^{-6}\%$, smaller than the claimed "floor" by seven orders of magnitude. The same failure occurs in the near-uniform regime of the actual meshes, where the masses vary by less than a factor of two and count-quantization alone admits deviations well below $\max_i v_i/\bar{A}$.

What survives, and what the project actually uses, is weaker and still useful: $\max_i v_i/\bar{A}$ is the natural *quantum* of the problem — a single vertex is indivisible, so balance is adjusted in steps of that size — and it is therefore a sensible *scale* against which to state acceptance bars. It is a calibration convention, not a bound. *What the shipped subgradient method attains in practice is an empirical question, not a consequence of Proposition 6.3.*

Remark 6.6 (Measured values)**.** $\max_i v_i/\bar{A}$ is 1.011% at $V = 47{,}488$, $N = 300$; 0.421% at $V = 114{,}144$, $N = 300$; 0.140% at $V = 114{,}144$, $N = 100$; and 0.561% at $V = 114{,}144$, $N = 400$. The floor scales as N/V — inversely with vertices per cell — for the same reason (7) does.

7 Connectivity repair

The offsets control area but, by the second bullet of §6.3, not connectivity. A second pass restores it. It operates on labels only.

Stage 1 — island absorption. For each cell, compute the connected components of its territory in the mesh edge graph. *Every* non-largest component — regardless of size — is relabelled to the neighbouring cell with which it shares the longest boundary. (The `fragment_rel_threshold` parameter governs only what the validity *gates* report as a genuine stray; absorption itself applies no threshold, and in the $N = 300$ run it absorbed six components, two of them below that reporting threshold.) This step is unconditionally safe: a component adjacent to cell c can always be merged into c without disconnecting c, since the merged set is connected through the shared boundary. On exit every cell is connected (or empty, which is reported).

Stage 2 — area restoration by boundary transfer. Absorption perturbs the areas, so equal areas are restored by moving *single boundary vertices* from richer cells to poorer ones. A move of vertex i from donor d to receiver r is admitted only if

$$T_d - T_r \;>\; v_i, \tag{14}$$

which guarantees that the transfer strictly reduces the *quadratic* imbalance $\sum_k (T_k - \bar{A})^2$, by $2v_i(T_d - T_r - v_i) > 0$. (It does *not* guarantee a strict decrease of $|T_d - \bar{A}| + |T_r - \bar{A}|$: with $\bar{A} = 10$, $T_d = 10$, $T_r = 4$, $v_i = 5$ the move is admitted and that sum is unchanged.) The move is taken only if $\Omega_d \setminus \{i\}$ remains connected — an exact articulation test, cheap because it is local to one cell. Candidates are ranked by the score margin $s_{ir} - s_{id}$, so the least costly transfers are taken first.

Remark 7.1 (What is and is not guaranteed)**.** Connectivity holds on exit unconditionally, since every move is gated by the articulation test. The second invariant — no admissible improving move remaining — holds *unless the sweep cap was reached*, which the implementation reports as `hit_sweep_cap` and which has occurred in practice. Absent the cap, termination is guaranteed: each move strictly decreases $\sum_k (T_k - \bar{A})^2$, the label space is finite, so no state can repeat. What is **not** guaranteed is that the exit state attains the granularity floor (13): stage 2 is a greedy local search over single-vertex moves, and a configuration can be locally optimal without being globally balanced, particularly when (14) blocks every candidate. The implementation therefore reports its own strain — number of moves, moves blocked by the connectivity test, sweeps used, and whether the sweep cap was reached — so that a run beyond what repair can fix says so rather than returning a degraded partition silently.

Code: the two-stage readout

`src/partition/balanced_readout.py`: `apply_balanced_readout(...)` orchestrates `solve_dual_offsets` → `reassign_islands` → `rebalance_boundaries`; `_stays_connected` is the articulation test and `build_vertex_adjacency` the edge graph, built from `faces` so that it respects the true surface topology (e.g. the torus periodic wrap) rather than a flat parametrisation. CLI: `scripts/balanced_readout.py`.

8 Relation to power diagrams

With scores $s_{ik} = -d^2(i, z_k)$ for sites z_k, (10) reads $\omega(i) = \arg\max_k(-d^2(i, z_k) + \psi_k) = \arg\min_k(d^2(i, z_k) - \psi_k)$, which is precisely a *power diagram* (Laguerre tessellation) with weights ψ_k. That the weights can be chosen to prescribe the cell capacities is Aurenhammer, Hoffmann and Aronov's Minkowski-type theorem [2]; modern solvers for the continuous version are given by

Mérigot [7] and Kitagawa, Mérigot and Thibert [6], and the construction is used for blue-noise sampling by de Goes *et al.* [5].

The balanced readout is the same algebra applied to a different score. Two differences are worth recording:

1. Our scores are $\log u$, a *relaxation output*, not a distance. There are no sites, and no geometric structure is assumed — only that the score is additive in ψ.

2. Our problem is finite-to-finite, so the damped Newton methods of [6], which exploit the smoothness of the semi-discrete objective in the continuous-source setting, do not transfer directly; D in (9) is piecewise linear, hence nonsmooth. Subgradient descent is used instead. Entropic regularisation [4] is an alternative that was implemented and measured for this problem, and lost to subgradient descent on five of six score matrices (`docs/experiments/07-phase0-shared-harness/`).

Remark 8.1 (We are not aware of this being reported)**.** Applying a capacity-constrained assignment as a *readout correction for a* Γ*-convergence relaxation* — as opposed to as a construction in its own right — is not something we have found in the literature. Stated as a population claim, the corpus searched is this project's bibliography plus two full-text readings; that is not a systematic review, and the claim is phrased accordingly.

9 Measured effect

Recorded here for completeness; the study that measures them belongs in `docs/experiments/`, which does not yet contain one for this stage.

torus, $N = 300$, $V = 47{,}488$	source	after readout	
imbalanced cells (gate 5%)	10	**0**	
worst cell deviation	36.15%	**1.63%**	floor 1.011%
disconnected cells	2	**0**	
label-boundary length	189.73	193.30	+1.88%
wall time	—	$\approx 17\,\mathrm{s}$	

The +1.88% boundary cost is the price of the correction: territory is moved to meet the area target, and moved territory has a longer border. It is *expected* that Phase 2 recovers most of it, but this is not measured and cannot be with the data available: the uncorrected source labelling fails validity and cannot be fed to Phase 2, so no counterfactual exists.

Summary table

Quantity	Status	Where	Implementation
Territory–mass identity (5)	derived, exact	§3	—
Band fraction (7)	derived, scaling	§4	—
Worst-of-N growth	**model, unmeasured**	§4	—
Disconnection admissible	derived	§5.1	—
Disconnection *mechanism*	**hypothesis, untraced**	§5	—
Dual (9), subgradient (11)	derived	§6	`solve_dual_offsets`
Nested growth (locality)	derived	§6.3	—
Exact balance unattainable (13)	derived; the vertex mass is a *scale*, **not** a floor	§6	`vertex_granularity` (in a
Island absorption	derived, safe	§7	`reassign_islands`
Boundary transfer	greedy; invariants only	§7	`rebalance_boundaries`

References

[1] Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, Upper Saddle River, NJ, 1993. Transportation problem, basic feasible solutions and the integrality property (which requires integer supplies – see `docs/math/09-balanced-readout`). Publisher and authors verified from the copy held; year 1993 is the standard edition and was not read off the title page.

[2] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998.

[3] Beniamin Bogosel and Édouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 26(4):496–508, 2017. THE foundational reference for this project: Phase 1 (Γ-convergence relaxation) and Phase 2 (direct perimeter minimization). Distinct from `bogosel2021longest`, which is the same authors on the OUTER problem and is cited only in the literature section of experiment 08 – do not conflate them. read `year = 2023` with no volume or pages, under the key `bogosel2023partitions`. The rest of the repository already said 2017 (`src/surfaces/double_torus.py`, `banchoff_chmutov.py`, `docs/plans/MESH_DEGENERACY_AND_NEEDLE_TRIANGLES.md`); the bibliography was the outlier. Taylor & Francis lists this as Experimental Mathematics *volume 26 (2017), issue 4*, and the article page gives "Published online: 04 Oct 2016". Crossref carries both dates and its automatic BibTeX transform emits `year = 2016`; the DOI also encodes 2016. We cite the ISSUE year, 2017. A venue preferring online-first dating would write 2016; both refer to this same article, DOI 10.1080/10586458.2016.1223570, arXiv:1606.02873.

[4] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.

[5] Fernando de Goes, Katherine Breeden, Victor Ostromoukhov, and Mathieu Desbrun. Blue noise through optimal transport. *ACM Transactions on Graphics*, 31(6):171:1–171:11, 2012.

[6] Jun Kitagawa, Quentin Mérigot, and Boris Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society*, 21(9):2603–2651, 2019.

[7] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5):1583–1592, 2011.

[8] Luciano Modica. The gradient theory of phase transitions and the minimal interface criterion. *Archive for Rational Mechanics and Analysis*, 98(2):123–142, 1987. Classical Γ-convergence result for the Modica–Mortola functional; source of the interface constant $c_W = 2\int_0^1 \sqrt{W}$ used in the Phase 1 energy.

[9] Luciano Modica and Stefano Mortola. Un esempio di γ-convergenza. *Bollettino dell'Unione Matematica Italiana B (5)*, 14:285–299, 1977. The original Γ-convergence example behind the Phase 1 relaxation energy.

[10] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019. Held as `docs/papers/1803.00567v4.pdf` (arXiv v4, 18 Mar 2020). Standard modern reference for the semi-discrete dual and its solvers. Journal, volume 11, number 5–6, pages 355–607 and year 2019 are all verified: the arXiv copy carries its own FnT citation block.

[11] Cédric Villani. *Topics in Optimal Transportation*, volume 58 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2003. Kantorovich duality. Series and volume verified from the copy held in `docs/papers/`.